



Effects of the Oxygen depletion in FLASH irradiation investigated through Geant4-DNA toolkit

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ARTICLE INFO

Keywords:

FLASH-RT

Monte Carlo simulation

Geant4-DNA

Oxygen enhancement ratio

Oxygen depletion

ABSTRACT

FLASH radiotherapy (or FLASH-RT) is a novel radiotherapy technology consisting of radiation delivery at dose rates several orders of magnitude higher ($\geq 40 \text{ Gy/s}$) than the currently used in conventional clinical radiotherapy. Many recent in-vivo preclinical studies indicate that FLASH-RT can greatly spare healthy tissues while maintaining unchanged tumour control. The generally acknowledged, though not entirely substantiated, explanation for the FLASH effect relates to the oxygen depletion that occurs after the radiation passage. On the other hand, oxygen depletion or, more in general, oxygen-related effects are still not fully clarified. Different research groups carried out the Monte Carlo simulations of electron and proton irradiations in oxygenated water to evaluate the oxygen-concentration-related effects at the cell-scale level. We analysed and compared the simulation results of the oxygen effect under the FLASH condition (considering the time-dependent G-values and the oxygen enhancement ratio-weighted dose) we obtained with GEANT4-DNA against TRAX-CHEM code results in the literature. Our results indicate that oxygen depletion has a negligible effect on radiosensitivity via oxygen enhancement, showing a close agreement with the TRAX-CHEM code. The conclusion is that the Geant4-DNA toolkit can be a valid instrument to study the FLASH effect.

1. Introduction

Various experimental studies on animal models (Bourhis et al., 2019; Montay-Gruel et al., 2017; Vozenin et al., 2019; Beyreuther et al., 2019; Favaudon et al., 2014) recently indicated that radiation therapy (RT) with an ultrahigh-dose rate (UHDR; $\geq 40 \text{ Gy/s}$), remarkably increases the radioresistance of normal tissues, while the tumour control remains similar to conventional treatment. This enhanced radioresistance of the normal tissues could permit the realization treatments with a significative dose escalation to the tumour regions.

The FLASH effect seems to be connected not only to the peak dose-rate values but also to other parameters that seem to play a decisive role in the observed sparing effects. In particular, based on Table 1, parameters such as mean dose rate, dose per pulse and total irradiation time seem essential for the pulsed delivery of FLASH-RT and should be

considered (Durante et al., 2017; Buonanno et al., 2019; Wilson et al., 2020; Hughes et al., 2020).

Some research identified the oxygenation mechanisms (Spitz et al., 2020) as potential reasons for the sparing of normal tissue in FLASH conditions; nonetheless, the physical-biological explanation of the FLASH phenomenon is not fully understood. The degree of oxygenation is one of the elements that significantly contributes to the destruction of biological systems (Sonntag, 1987; Boscolo et al., 2018) through indirect damages. The indirect damage, described in detail elsewhere (Elgazzar and Kazem, 2006; Meylan et al., 2017; Lampe, 2017), is caused by chemical reactions among water radiolysis-induced free radicals (highly reactive molecules) and atoms and molecules in cells, especially DNA molecules.

Compared to anoxic cells, well-oxygenated regions are more sensitive to radiation and, consequently, respond better to therapeutic

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<https://doi.org/10.1016/j.radphyschem.2023.111184>

Received 4 February 2023; Received in revised form 23 June 2023; Accepted 24 July 2023

Available online 26 July 2023

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Table 1
Pulsed FLASH-RT delivery parameters.

Dose per pulse (pulse duration × Dose rate within the pulse)	≥ 1 Gy
Total dose (Dose per pulse × number of pulse)	≥ 10 Gy
Total delivery time	≤ 0.1 s
Mean dose rate($\frac{\text{Total dose}}{\text{Total delivery time}}$)	≥ 100 $\frac{\text{Gy}}{\text{s}}$

modalities (Tinganelli et al., 2015; Hall and Giaccia, 2012). Indeed, the presence of oxygen primarily increases the production of reactive oxygen species (ROS), causing more cells to be damaged that can be permanently “fixed” by molecular oxygen and make DNA damage unreparable. This phenomenon is most commonly described by the oxygen-fixation hypothesis (Grimes and Partridge, 2015). This increased sensitivity to radiation is known as the oxygen enhancement ratio (OER), which is defined as the ratio between the corresponding doses required at different oxygenation levels in anoxia and in a particular oxygenated condition (P_{O_2}) to obtain the same biological effect (Tinganelli et al., 2015; Hall and Giaccia, 2012):

$$\text{OER}(P_{O_2}) = \frac{D_{\text{anoxia}}}{D_{P_{O_2}}} \Bigg|_{\text{same biologic effect}} \quad (1)$$

The tissue in UHDR radiation is exposed for a short time. The first part of the dose in the pulse is used to deplete the oxygen, to allow the rest of the pulse to irradiate in anoxic conditions. Since the indirect damage is caused by free radicals depending on the molecular oxygen abundance, the indirect damage will be reduced and, therefore, the total DNA damage (Espen et al., 2020).

In this work, the effect of UHDR radiation, characterizing the FLASH condition, on the chemical species and, consequently, the possible effect of oxygen depletion, by considering the OER factor, has been simulated via the Geant4-DNA toolkit, used to investigate the correlation between molecular oxygen concentration and the indirect damages. During the UHDR irradiation, water-dissolved molecular oxygen is consumed according to interactions between different chemical species (see 2.1). In the FLASH oxygen depletion hypothesis, transient anoxia is created, boosting the radiation resistance, since re-diffusion of oxygen to the irradiated target, occurring in a time scale of milliseconds, may be overlooked during this brief irradiation duration (microsecond order). Through Geant4-DNA, this oxygen consumption and the ensuing indirect damage may be quantified.

Monte Carlo approach was already used by several groups to simulate the oxygenated water with different oxygen concentrations in electron and proton irradiations (Stepan and Davidkova, 2008; Coliaux et al., 2015; Boscolo et al., 2020; Lai et al., 2021); Stepan and Davidkova (2008) evaluated the effects of oxygen concentration on the radiolysis process of water and yields of primary DNA damages induced by ionizing radiation using the code TRIOL. Coliaux et al. (2015) studied ion radiation-induced free radical production in pure and oxygenated water and analysed the time-dependent G-values (the number of reactant molecules consumed or produced per 100 eV energy deposition Breen et al., 1995) of chemical species generated by projectiles with different LET (linear energy transfer) values.

Boscolo et al. (2020), presented Monte Carlo particle track structure code TRAX-CHEM to calculate the G-values of the main chemical species in water, with different oxygen concentrations and various irradiation conditions, as did Lai et al. (2021) with a GPU-based microscopic Monte Carlo code.

Additionally, Boscolo et al. (2021), demonstrated, using the radiation chemical models and oxygen enhancement factor used in this study, that oxygen depletion is not supported as the primary mechanism for the tissue-sparing effect in UHDR radiation. This was done through studies on the chemical track evolution of 1 MeV electrons in oxygenated water with “TRAX-CHEM”.

We exploited the power of the Geant4-DNA libraries (Incerti et al., 2010b,a; Bernal et al., 2015; Incerti et al., 2018; Shin et al., 2021),

included in the Geant4 Monte Carlo toolkit (Karamitros et al., 2011), to estimate the influence of oxygen concentration in the biological effects of particle irradiations. Firstly the time-dependent G-values were calculated for some chemical species in water with different oxygen concentrations for proton and electron irradiation. Then, the OER-weighted dose, under dynamic oxygen pressure, was calculated and, at last, compared with conventional irradiation at constant OER to evaluate the effect of high dose rate irradiation on chemical species. The results obtained for G-values and OER-weighted dose were compared with what was obtained by Boscolo et al. (2021) using TRAX-CHEM code.

2. Materials and methods

2.1. Geant4-DNA and chemical reactions implementation

To evaluate the role of oxygen concentration and depletion in the mechanism leading to the FLASH effect under electron and proton irradiation, the Geant4-DNA (Incerti et al., 2010b,a; Bernal et al., 2015; Incerti et al., 2018; Shin et al., 2021; Kyriakou et al., 2022a) particle track structure code was adopted.

Geant4-DNA is a set of libraries, fully included in Geant4 (Allison et al., 2006, 2016; Agostinelli et al., 2003) (“GEometry AND Tracking”) Monte Carlo toolkit, able to simulate the transport and the energy depositions of ionizing radiation in biological materials modelled as water at a nanoscopic scale and down to very low values (10 eV scale) of the incident particles energy. This toolkit can simulate the physical, physico-chemical, and chemical stages of the radiolysis of water in microseconds time scale after irradiation. It was shown that by combining this simulation with the ability to simplify the modelling of various biological geometries, such as bacterial cells and human fibroblasts, we can predict direct and indirect DNA damages induced by irradiation (Incerti et al., 2010b,a; Bernal et al., 2015; Incerti et al., 2018).

To realize our study, we based our simulation on the “scavenger” example, released with version 11.0.0 of Geant4 (Chappuis et al., 2023). We hence modified the code of this example to reproduce the conditions needed for our comparative study, in particular for the proton irradiation case.

For the range of energies applied in this study, the physics constructors *G4EmDNAPhysics_option2* and *EmDNAChemistry* from the “scavenger” example were used to simulate the physical and chemical stages of the particle interactions, respectively (Kyriakou et al., 2017, 2015).

The energy transfer caused by particle radiation in a water target happens instantly, with times on the order of femtoseconds [fs], and causes the synthesis of water radiolysis species as a result of the ionization and excitation processes in the Geant4-DNA physical stage. After this stage, the pre-chemical stage starts with events such as thermalization, the solvation of sub-excitation electrons, electronic recombination and electronic hole migration, occurring with time scales of the picoseconds [ps]. At last, there is the chemical stage where radio-induced species diffuse and react with each other.

Besides, the generated chemical species can dissociate into free radicals because of specific algorithms available in the chemistry module (Shin et al., 2021; Karamitros et al., 2011). The Geant4-DNA chemistry module includes different chemical information, such as the dissociation scheme of water molecules, species diffusion models and the molecular species details (mass, charge, reaction rate and diffusion constants). The chemical phase starts when all physical tracks were processed, and in this work, the chemical reactions are implemented using the independent reaction time (IRT) approach (Mendez et al., 2020). Several factors play a role in the occurrence of the various chemical reactions, including reaction rate constants, species diffusion models, and the presence of competing reactions.

EmDNAChemistry constructor used in this example uses an external file defining the possible chemical reactions and the oxygen

Table 2
Reactions among radio-induced species.

Reactions	Rate(1/s/(mol/l)) or (1/s)	Type
$e_{aq}^- + e_{aq}^- + [H_2O] + [H_2O] \rightarrow H_2 + OH^- + OH^-$	6.36e9	type_III
$e_{aq}^- + OH \rightarrow OH^-$	2.95e10	type_II
$e_{aq}^- + H + [H_2O] \rightarrow H_2 + OH^-$	2.50e10	type_I
$e_{aq}^- + H_3O^+ \rightarrow H + [H_2O]$	2.11e10	type_IV
$e_{aq}^- + H_2O_2 \rightarrow OH^- + OH$	1.10e10	type_II
$OH + OH \rightarrow H_2O_2$	5.50e9	type_II
$OH + H \rightarrow [H_2O]$	1.55e10	type_II
$H + H \rightarrow H_2$	5.03e9	type_I
$H_3O^+ + OH^- \rightarrow [H_2O] + [H_2O]$	1.13e11	type_III
$e_{aq}^- + [H_2] \rightarrow O_2^{\cdot -}$	1.74e10	type_VI
$H + [H_2] \rightarrow HH_2^{\cdot}$	2.10e10	type_VI
$O^- + [H_2] \rightarrow O_3^{\cdot -}$	3.70e9	type_VI

^aThe reactions are those involving the scavenger (i.e. the O_2 molecule) (Chappuis et al., 2023).

concentration. The classification of chemical reactions into six groups (type I to VI) (Pimblott and LaVerne, 1997) based on the diffusion theory (Park and Agmon, 2008; Mendez et al., 2020) is specified. The reaction probability is determined from the reaction times (the rate constants of chemical reactions), calculated via functions of the diffusion equation (Plante et al., 2017). Reactions used in this study, with their reaction rate and type, are listed in Table 2.

We aim to study how chemical species, more specifically free radicals induced by water radiolysis, can change the concentration of dissolved oxygen molecules in water. Solvated electrons (e_{aq}^-) and hydrogen radicals ($H^{\cdot}$) are two of the most abundant of these radiation-induced free radicals, and they can interact with oxygen molecules dissolved in water based on the following two reactions:



Reactions (2) and (3) show how dissolved oxygen molecules can be consumed by e_{aq}^- and $H^{\cdot}$.

We assumed and used dissolved oxygen molecules as a continuum backdrop, with a constant diffusion rate of zero and a homogeneous distribution in water.

To simulate the interaction of dissolved oxygen molecules with the chemical species induced by water radiolysis, the “reactions type VI” (see Table 2) were adopted (Chappuis et al., 2023). The reactions type VI are defined as “reactions with the background”; this means that the reactants can interact as the first-order reactions (reactions containing just one reactant) (Kyriakou et al., 2017) with the continuum approach (Green et al., 1990) and can be defined as an interaction with a “scavenger” (Plante et al., 2017). The “scavenger” is a chemical substance added to a mixture to remove or deactivate impurities and unwanted reaction products. In our case, the scavenger is the oxygen molecules (O_2) and was modelled as a reservoir with fixed concentration. Based on the reactions (2) and (3), O_2 interacts with e_{aq}^- and $H^{\cdot}$, consuming them (Colliaux et al., 2011). These reactions show that e_{aq}^- and $H^{\cdot}$ are scavenged by molecular oxygen. Therefore, the conversions of molecular oxygen into superoxide anions ($O_2^{\cdot -}$) and perhydroxyl ($HO_2^{\cdot}$) (reactions (2) and (3)) are interactions of major importance in our study (Beuve et al., 2015).

2.2. Oxygen concentration implemented in Geant4-DNA

To simulate the variation of oxygen concentration inside a biological tissue, different concentrations of dissolved molecular oxygen were added to the water in the simulations (see Section 2.3). In this study, the oxygen concentration is expressed in terms of oxygen partial pressure. The conversion from partial pressure to concentration expressed in the M unit is given by Henry's law (Wenger et al., 2015).

According to this law, the oxygen concentration (C_s) in the water box is proportional to the partial oxygen pressure (P_{O_2}):

$$C_s = K_H \times P_{O_2} \quad (4)$$

where K_H is Henry's constant which is proportional to the gas solubility in water, whose value for O_2 is $1.0264 \times 10^{-5} \text{ mol}/(\text{m}^3 \text{ Pa})$ at 37°C . For example, for a well-oxygenated target with oxygen partial pressure in per cent $P_{O_2}(\%)$ of 21%, the corresponding oxygen concentration (C_s) is 218 μM .

We considered molecular oxygen concentrations ranging from $P_{O_2}(\%) = 0\%$ (representing an anoxic condition) to well-oxygenated targets ($P_{O_2}(\%) = 21\%$).

2.3. Geometrical set-up and primary beam characteristics

In the physical stage, the box material is water, while for the chemical one, it is water with the addition of a specific oxygen concentration. This concentration is supposed to have no influence on the physical stage, and the molecular oxygen is homogeneously distributed in the target and treated as a continuum.

Two geometrical set-ups were simulated, one for the electron case and the other for the proton one.

2.3.1. Electrons

For our simulation with electron beams, the primary particles, with a kinetic energy of 0.5 and 1 MeV, were shot from a point source located at the centre of a 1 km^3 water “box” (Fig. 1A). Since the electron deposits its energy sparsely, the water box was modelled large enough to stop all primary and secondary particles produced inside it (the water box represents a “pseudo-infinite” homogeneous region). In this infinite volume, the energy lost by the primary particles equals the deposited one, as all secondary particles are reduced to thermal energy. Such a vast volume needs a lot of CPU time to simulate. In order to avoid this issue, we only took into account a portion of the track (a track segment) that was related to a range of the deposited energy for all calculations of the relevant quantities of this study (see Section 2.4), in accordance with Ramos-Mendez et al. (2018), and Uehara and Nikjoo (2006). It means the primary electron was stopped as soon as its accumulated energy loss along the track exceeded a threshold of energy deposition (E_{thr}). Indeed, track segment simulations limit the computation of G-values around the segment of the track where the particle keeps its energy close to the primary energy. The choice of E_{thr} , and so consequently of the track segment length, is done by mediating between the need for appropriate statistics on the number of radiolytic species and the necessity of limiting the CPU time.

2.3.2. Protons

For protons, being densely ionizing particles, a small volume is enough to obtain a good compromise between statistics and CPU time. In this case, the simulated geometry consisted of a $10 \times 10 \times 10 \mu\text{m}^3$ cubic water “target” (the sensitive volume), where all relevant quantities (see Section 2.4) were calculated, located inside another 1 km^3 water box. This geometry was irradiated with a monodirectional and monoenergetic 90 MeV proton point source placed in the centre of the target (see 1B).

2.4. Computed quantities

Geant4-DNA pre-chemical and chemical modules are devoted to simulating the production of radiochemical reactive species, their diffusion, and their interaction mechanisms in the target after the physical stage has happened. We calculated parameters such as oxygen consumption and time-dependent G-values of different chemical species produced in the oxygenated water following electron and proton irradiation.

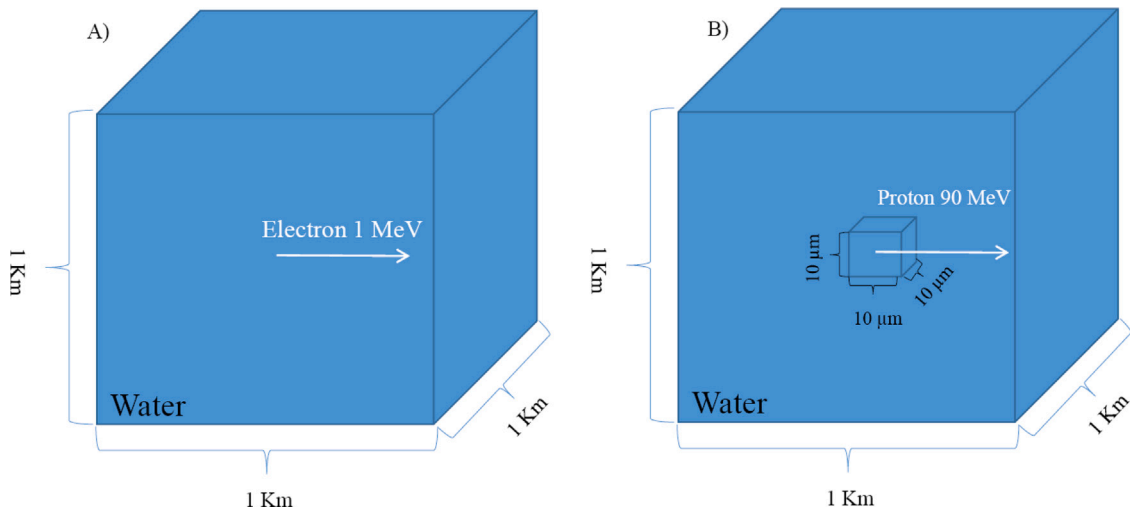


Fig. 1. Set-up scheme for electrons (A) and protons (B).

We studied the dependence of the FLASH effect on the oxygenated target, which is the so-called oxygen depletion hypothesis, considered as one of the possible mechanisms to explain the FLASH effect (Petersson et al., 2020).

Oxygen consumption was calculated as a function of dose, and then the OER-weighted dose was calculated under dynamic oxygenation conditions. Finally, to evaluate the FLASH effect, the results obtained were compared with conventional irradiation conditions at constant OER.

2.4.1. G-values in the water at different oxygen concentrations

Utilizing the capabilities of Geant4-DNA, the energy deposition and the number of reactive species created (or lost) in the target along the chemical evolution period were estimated in order to extract the radiochemical yield based on the time-dependent G-value definition. The time-dependent G-value parameter allows us to figure out the mechanisms of reaction dynamics within the chemical track.

The concentrations of dissolved molecular oxygen were assumed to range from anoxic ($P_{O_2} = 0\%$) and anoxic ($P_{O_2} = 1\%–3\%$) conditions for typical tumours to physioxia conditions ($P_{O_2} = 7\%$) and well-oxygenated targets ($P_{O_2} = 21\%$).

Under these conditions, track segment G-values were calculated for all major chemical species for proton and electron irradiation.

2.4.2. OER -weighted dose under varying oxygenation

We concentrated our work on the chemical stage and the OER to determine the changing oxygen consumption rate in order to replicate the impact of oxygen depletion during the brief irradiation time.

The simulation of oxygen consumption, and its correlation, expressed by the OER, with the radiobiological effect of oxygen-enhanced radiosensitivity, was used to investigate the tissue-sparing effect during transient anoxia of oxygenated target in dependence of dose rate. As many references suggest (Wilson et al., 2020; Esplen et al., 2020; Gray et al., 1953; Cao et al., 2021), radiation-induced anoxia decreases radiosensitivity, and, as mentioned earlier, this sensitivity can be quantified by the OER factor (Eq. (1)).

Dose-rate is an important parameter in the FLASH effect, but in the present work, due to the existence of limitation in dose rate simulation using the Geant4 toolkit (particles are considered one by one), the oxygen depletion was weighted as a function of dose. In the one single-track simulation, to satisfy the condition that an extremely short irradiation time (on the microsecond scale) does not allow the target to be reoxygenated, we considered the irradiated water box sealed (see Section 2.3). This maximizes the amount of oxygen depletion. In fact, an increase in dose rate increases the inter-track interaction, thereby

increasing the recombination between radicals, and, as a result, the interactions between radicals and O_2 are reduced (Boscolo et al., 2021). Hence, we can assess the upper limit for oxygen depletion by ignoring the dose rate and quantifying it by the OER index.

In the first step, the time-dependent G-values of acid perhydroxyl ($HO_2^\bullet$) and superoxide ($O_2^{\bullet-}$) expressed in M/Gy were calculated using the Geant4-DNA simulation, then, according to the numerical equation below (5), the dose-dependent amount of $O_2^{\bullet-}$ plus $HO_2^\bullet$ was calculated:

$$[O_2^{\bullet-} + HO_2^\bullet] = \int G_{O_2^{\bullet-}+HO_2^\bullet}([O_2](D)) dD. \quad (5)$$

$$[O_2] = [O_2]_{ini} - \int G_{-O_2}([O_2](D)) dD. \quad (6)$$

Since an acid–base pair equilibrium is quickly established in the biological medium, the sum of $O_2^{\bullet-}$ and $HO_2^\bullet$ can be assumed equal to the amount of oxygen consumed through reactions (2) and (3). In the next step, the dose weighted by OER (D_{OER}) under dynamic oxygenation in the sealed target ($D_{OER,DYN}$) was calculated and then compared with D_{OER} obtained by the initial constant oxygen concentration ($[O_2]_{ini}$) under conventional irradiation ($D_{OER,CONV}$), as proposed by (Boscolo et al., 2021):

$$D_{OER,DYN} = \int OER([O_2](D)) dD \quad (7)$$

$$D_{OER,CON} = D_{OER}([O_2]_{ini}). \quad (8)$$

The parametrized OER(pO_2), as suggested by Grimes and Partridge (2015), is:

$$OER(pO_2) = 1 + (\Phi_0/\Phi_D)(1 - \exp(-\varphi pO_2)), \quad (9)$$

$$(\Phi_0/\Phi_D) = 1.63, \quad \varphi = 0.2567)$$

where the constants ($\Phi_0/\Phi_D, \varphi$) were fitted for a specific biological endpoint (cell killing).

3. Results and discussion

In this section, we present all the results of our study, which we list for simplicity in Table 3:

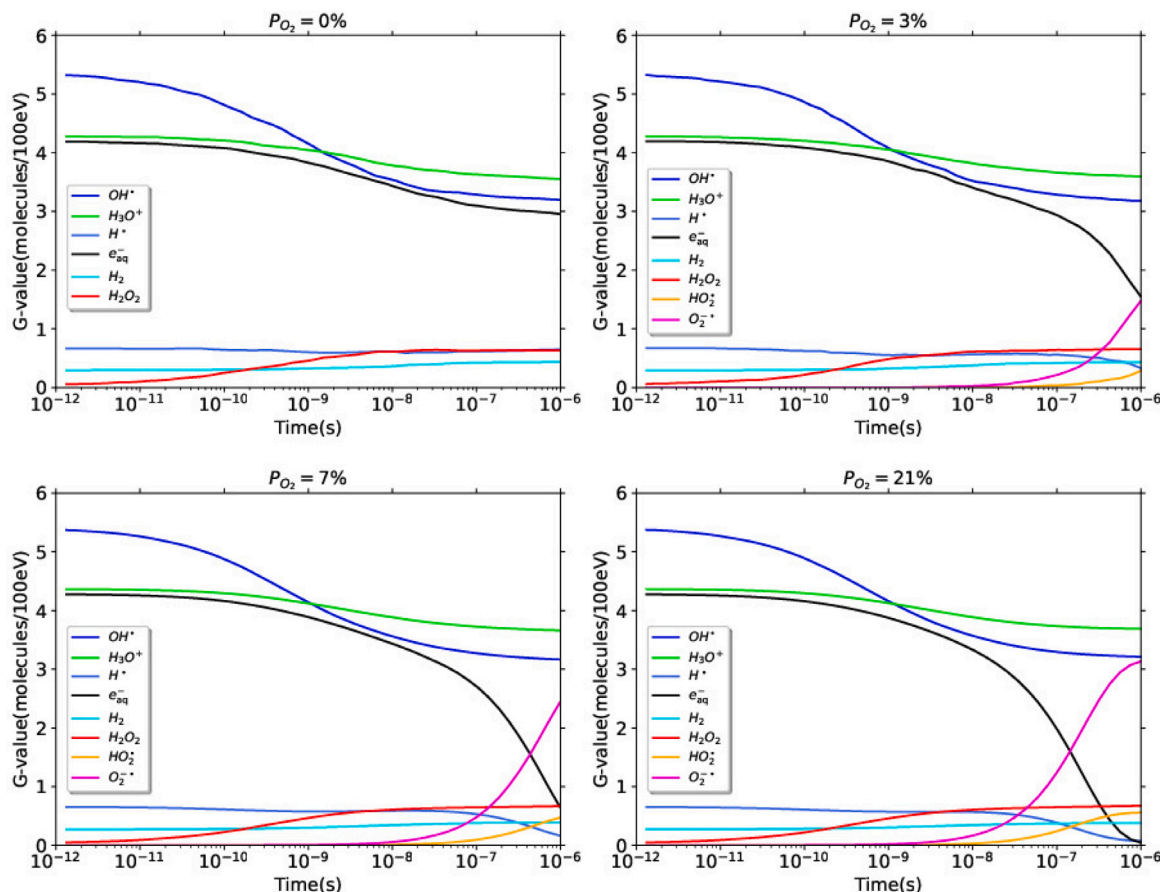
3.1. G-values in the water at different oxygen concentrations

Through the simulation of time-dependent G-values of various chemical species formed in oxygenated water, the mechanism of reaction dynamics inside the chemical stage, after irradiation with 90 MeV

Table 3

List of all the results presented in this paper.

Par.	Fig.	Content
3.1	2	G-values of different chemical species produced by 90 MeV protons with different oxygen concentrations.
3.1	3	Comparison, with TRAX-CHEM code, for the G-values of different chemical species produced by 90 MeV protons with one oxygen concentration.
3.1	4	Comparison, with three different track-structure codes, for the G-value of e_{aq}^- , produced by 5 MeV protons with one oxygen concentration.
3.1	5	Comparison, with TRAX-CHEM code, for the G-values of different chemical species produced by 0.5 MeV electrons with one oxygen concentration.
3.2	6	Oxygen concentration vs. dose under FLASH condition with 1 MeV electrons for different initial oxygenations, in comparison with TRAX-CHEM code results.
3.2	7	Ratio of the OER weighted dose for FLASH on conventional irradiation with 1 MeV electrons for different initial oxygenation as a function of dose (at 1 μ s), in comparison with the TRAX-CHEM code results.

**Fig. 2.** Time-dependent G-values of the chemical species produced by 90 MeV proton irradiation for four different oxygen concentrations (0%, 3%, 7% and 21%).

proton and after irradiation with 0.5 MeV electron, has been explored. Fig. 2 shows the results for the 90 MeV protons case, with four different oxygen concentrations (0%, 3%, 7% and 21%). The time interval considered in this simulation is 10^{-12} s– 10^{-6} s.

As mentioned in Boscolo et al. (2018), the most abundant of the chemical species formed in the pre-chemical stage are OH^* , H_3O^+ , and e_{aq}^- , which have a maximum yield at the beginning of the chemical stage (in the time scale of picosecond) and, being involved in various chemical reactions (see Table 2), decrease during chemical evolution time, as we can see in Fig. 2. Simultaneously, H_2O_2 and H_2 molecules are also produced but at much lower concentration.

Looking at Fig. 2, we can observe that the presence of the dissolved oxygen molecule in the water does not affect the formation of radicals in the first nanosecond after irradiation. Because, in this time scale, the recombination process between radiolytic species in water occurs more likely than their diffusion and interaction with dissolved oxygen due to the high density of interaction sites.

For times larger than a few nanoseconds after the irradiation, various chemical species begin to diffuse more effectively, increasing the probability of radical interaction with dissolved oxygen. This probability increases with the increase of dissolved oxygen concentration. As described earlier, dissolved oxygen plays the role of scavenger for both species e_{aq}^- and H^* , leading to the formation of the products $O_2^{\cdot-}$ and HO_2^* , which increases in abundance as the oxygen concentration increases from 0% to 21%: this is shown clearly in the four plots of Fig. 2.

One microsecond after the irradiation was chosen as an arbitrary time at which the chemical interaction rates between the radiation-induced free radicals are assumed to be very low (Boscolo et al., 2020).

At this time, under well-oxygenated conditions ($pO_2 = 21\%$), complete depletion of e_{aq}^- and H^* can be obtained, as shown in the last plot of Fig. 2. At lower oxygen concentrations (first three plots of Fig. 2), the likelihood of interaction among the above-mentioned radiolytic species and dissolved molecular oxygen is lower, so their complete depletion does not occur.

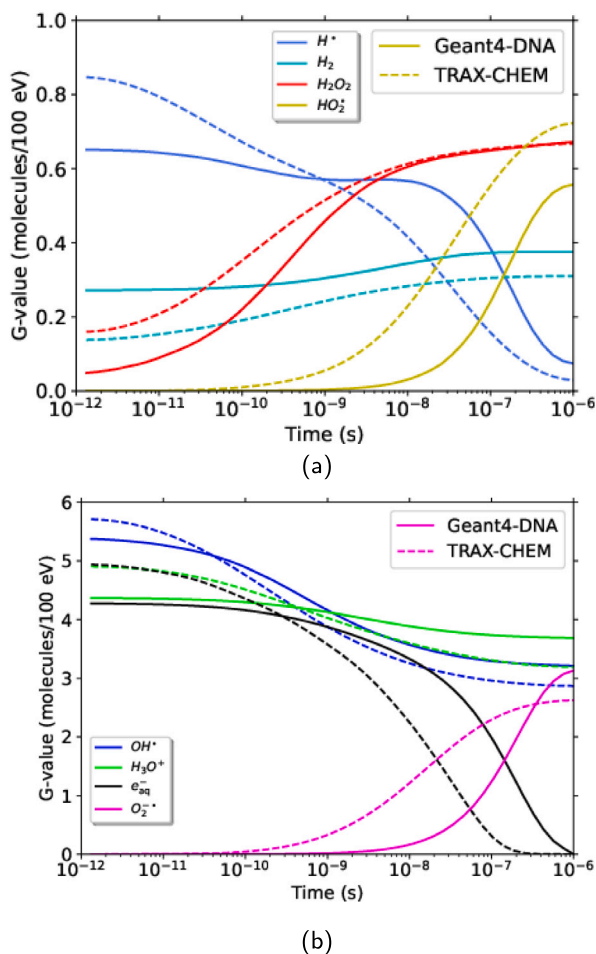


Fig. 3. The results obtained for the time-dependent G-values of the chemical species produced by 90 MeV proton irradiation for 21% oxygen concentration, in comparison with what published with TRAX-CHEM code (Boscolo et al., 2020).

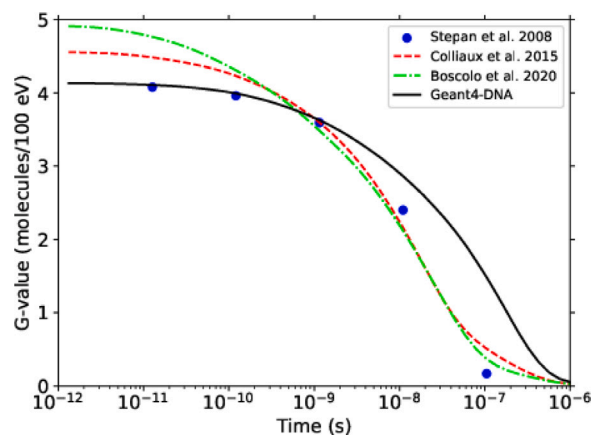


Fig. 4. Comparison among different track structure codes (TRIOL Stepan and Davidkova, 2008, TRACIRT Colliaux et al., 2015, TRAX-CHEM Boscolo et al., 2020 and Geant4-DNA (this work)) for the determination of time-dependent G-values for e_{aq}^- produced by 5 MeV protons in water with an oxygen concentration of 21%.

A comparison of the obtained results was performed with what was presented by other authors with different track structure codes. In Fig. 3, we compare the results we obtained with Geant4-DNA against

TRAX-CHEM results, presented by Boscolo et al. (2020), for the time-dependent G-values of various chemical species after 90 MeV protons irradiation in water and with an oxygen concentration of 21%.

In Fig. 4, the G-value of solvated electrons, produced by a 5 MeV proton irradiation in water with an oxygen concentration of 21%, obtained with Geant4-DNA, was compared to the results of three chemical track structure codes: beside TRAX-CHEM (Boscolo et al., 2020) also TRIOL (Stepan and Davidkova, 2008) and TRACIRT (Colliaux et al., 2015).

At last, in Fig. 5, we compared our results with those of TRAX-CHEM (Boscolo et al., 2020) for the time-dependent G-values of four chemical species (e_{aq}^- , $O_2^{\bullet-}$, $H^\bullet$ and $HO_2^\bullet$) after 0.5 MeV electron irradiation in water and with four different oxygen concentrations (0%, 2.5%, 5% and 21%). As can be seen in the figure, at an oxygen concentration of 21%, a saturation limit is reached. This limit is obtained when all e_{aq}^- and $H^\bullet$ are entirely depleted and production of both species $O_2^{\bullet-}$ and $HO_2^\bullet$ ceases. At other oxygen pressures (0%, 2.5%, 5%), not all e_{aq}^- and $H^\bullet$ are entirely scavenged by O_2 during one microsecond. The reaction kinetics of the whole track was hence limited to the lifetime of e_{aq}^- and $H^\bullet$ in the target.

Differences observed in the G-values obtained with those codes, shown in Figs. 3, 4 and 5, can be explained by the difference in the reaction types and dissociation probabilities assumed by each of them. Despite this, the codes are in good agreement for quantifying Oxygen depletion.

3.2. OER-weighted dose under dynamic oxygenation

Oxygen concentrations as a function of dose, under FLASH condition with 1 MeV electrons for various initial oxygenations ranging from physiologic normal tissues (5%, 7%) to anoxic tumours (1%), were studied. The results are shown in Fig. 6, together with the TRAX-CHEM results for comparison. Both codes show how oxygen consumption varies proportionally with the dose, and complete 'depletion' is not observed for any initial oxygen concentration considered.

In Fig. 7, we show the ratio between D_{OER} for FLASH ($D_{OER,FLASH} = D_{OER,DYN}$) and D_{OER} for conventional irradiation ($D_{OER,CON}$) as a function of dose, which can be used to estimate the sparing effect achievable with FLASH condition.

For normal tissue, an initial O_2 concentration of 5%–7% was assumed, corresponding to 65–91 μM at room temperature. As shown in Fig. 7, for those O_2 concentration values, the $D_{OER,FLASH}/D_{OER,CON}$ ratio is close to one for doses up to about 120 Gy and in particular for typical clinical doses between 2 and 10 Gy indicating that no sparing effect is achievable. For doses above 120 Gy, only a slight decrease in $D_{OER,FLASH}/D_{OER,CON}$ is observed. A more remarkable decrease in the $D_{OER,FLASH}/D_{OER,CON}$ ratio is found for lower oxygen concentrations (1%–0.5%), typical of anoxic tumours. As this plot shows, the ratio $D_{OER,FLASH}/D_{OER,CON}$ (at a given absorbed dose) is smaller at lower oxygen concentrations, and this means that more sparing occurs in anoxic rather than physiologic tissues.

As reported in many studies (Buonanno et al., 2019; Colangelo and Azzam, 2020), protons, with their unique properties (good depth penetration together with localized deposition of the majority of the dose, which allows for sparing normal tissues around the target), could, in principle, play a significant role in FLASH radiotherapy. However, that high LET radiation may reduce the enhancement effect of oxygen. This reduction is in terms of the vaster recombination of chemical species in a track of denser interaction sites, resulting in less production of two highly toxic species ($HO_2^\bullet$ and $O_2^{\bullet-}$) (Boscolo et al., 2020). So low-energy protons with their high LET would require much higher doses to achieve complete oxygen depletion compared to the low LET electrons studied here.

The effect of oxygen (expressed by the OER factor) is significant in the case of sparsely ionizing radiations such as electrons and does not have any useful role for dense ionizing radiations such as protons (Hall and Giaccia, 2012). For this reason, we chose to study the effect of oxygen only for electrons.

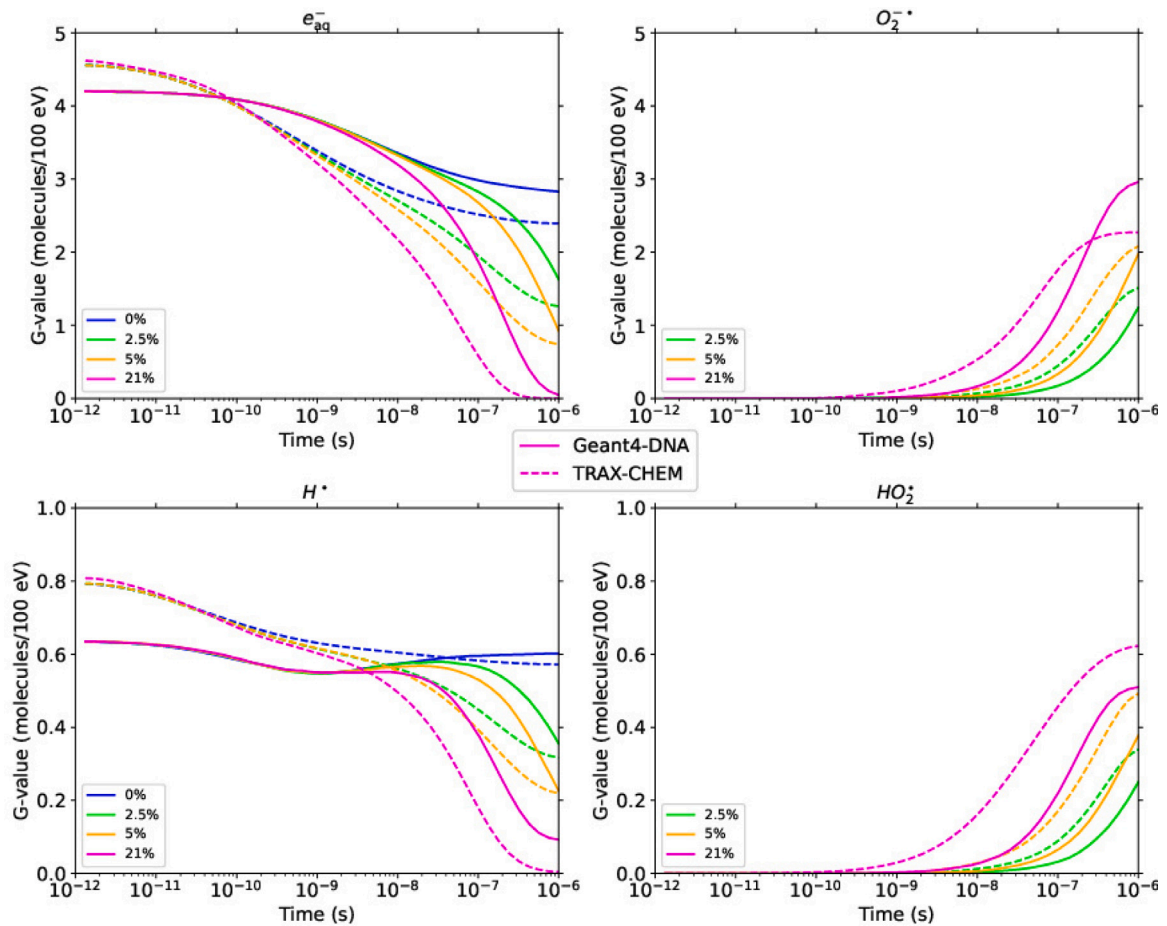


Fig. 5. Time-dependent G-values of the chemical species produced by 0.5 MeV electrons irradiation for different oxygen concentration, in comparison with what was published with TRAX-CHEM code (Boscolo et al., 2020).

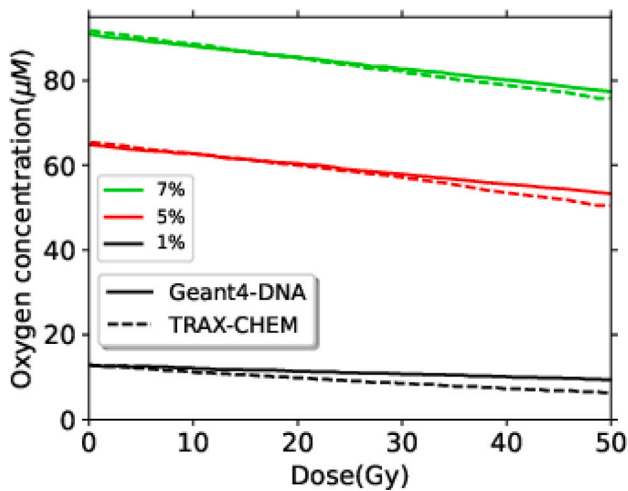


Fig. 6. Oxygen concentration as a function of dose under FLASH condition with 1 MeV electrons at various initial oxygenations, in comparison with the TRAX-CHEM code results (Boscolo et al., 2021).

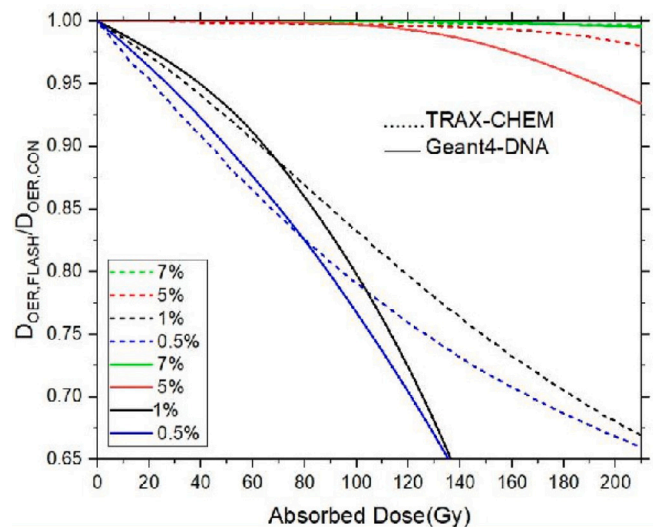


Fig. 7. The ratio of the OER-weighted dose for FLASH on conventional irradiation with 1 MeV electrons for different initial oxygenation as a function of dose (at 1 μ s), in comparison with the TRAX-CHEM code results (Boscolo et al., 2021).

4. Conclusion

The calculated G-values in the water target at different oxygen concentrations and the simulated OER-weighted dose under dynamic oxygenation were determined to study the oxygen depletion upon FLASH condition through the Geant4-DNA code. For the irradiation of

a homogeneous oxygenated water target, geometrical configurations with monoenergetic electrons (0.5 and 1 MeV) and protons (5 and 90 MeV) were taken into consideration. It has been possible to model the time-dependent G-values of the main radiolytic species at various

oxygen concentrations up to one microsecond after irradiation. Under oxygenated conditions, two toxic species ($O_2^{\bullet -}$ and $HO_2^{\bullet}$) are formed as reaction products of the interaction between e_{aq}^- and $H^{\bullet}$ with dissolved molecular oxygen. The amounts of $O_2^{\bullet -}$ and $HO_2^{\bullet}$ produced correspond to the ones of e_{aq}^- and $H^{\bullet}$ consumed. The extent to which these products are formed depends on the concentration of molecular oxygen in the water.

Using Geant4-DNA and considering the OER, which indicates, specifically for low LET radiation, the level of radiosensitivity in terms of oxygen, we have shown that oxygen depletion, as one of the hypotheses in FLASH condition, is not observed in the range of clinical doses. Moreover, complete oxygen depletion is not supported by the calculation model implemented in this work. Furthermore, we concluded that the sparing effect of FLASH condition is more effective in tumour tissue than in normal tissue. The results we obtained with Geant4-DNA were compared with those published by Boscolo et al. (2020, 2021), with the TRAX-CHEM code. A general agreement was observed in all cases analysed by these codes separately, with differences ascribable to the different reaction types, different list of reactions and dissociation probabilities assumed by each study. Despite those differences, the two studies come to the same conclusion regarding oxygen depletion as a principal mechanism for the FLASH effect.

Nevertheless, it is noteworthy to underline that relevant sources of uncertainties in the simulations we performed come from the physical and chemical parameters used in Geant4-DNA, as discussed in detail by Shin et al. (2019). Moreover, an additional source of uncertainty concerns the treatment of very low energy electrons (sub-100eV energies), which has a strong influence on the subsequent radiolysis. When we performed this study, the only available physics model for the energy range we treated was given by the physics constructors *G4EmDNAPhysics_option2*. Recently a more accurate physics model for low-energy electron transport in Geant4-DNA (Kyriakou et al., 2017, 2015) has been extended up to 1 MeV (Kyriakou et al., 2022b). It follows that even more accurate results could be obtained by repeating now the study with the new model mentioned above.

The conclusions we arrived at, compatible with those of Boscolo et al. (2020), are based on the analysis of the OER index. Nevertheless, Seco et al. (2022) proposed a mechanism by which tumour cells can modulate the oxygen concentration. Therefore, based on this study, the OER index may not be an ideal option for investigating the oxygen depletion phenomenon in correlation with the FLASH effect. Further studies are needed to confirm or deny this statement.

CRedit authorship contribution statement

Fateme Farokhi: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Babak Shirani:** Writing – review & editing. **Serena Fattori:** Supervision, Writing – review & editing. **Mohammad Ali Asgarian:** Reviewing. **Giacomo Cuttone:** Reviewing. **Sayed Bijan Jia:** Reviewing. **Giada Petringa:** Reviewing. **Alberto Sciuto:** Reviewing. **G.A. Pablo Cirrone:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This work was supported by Isfahan University, Iran. Moreover, it was performed in the framework of the MC-INFN (Monte Carlo at INFN) initiative, also funded by the Interdisciplinary Committee V of INFN. The authors want to thank F. Chappuis, L. Desorgher, S. Incerti, H. N. Tran, E. Scifoni, D. Boscolo and M. C. Fuss for all the valuable discussions and advice. F. Farokhi wants to thank V. R. Sharma (Dept. of Radiation Oncology, SoM-UMB, MD, USA) for all his support during the realization of this manuscript.

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